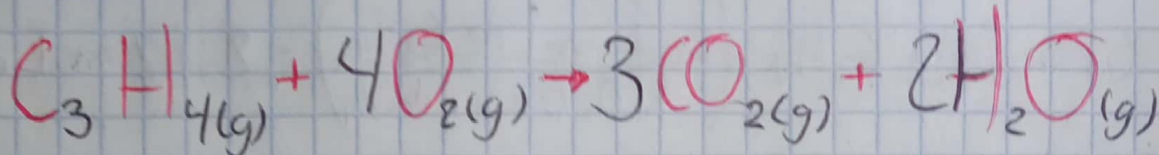


Examen

1.- Calcule cuanto calor se desprenderá si se hacen reaccionar 325 g de C_3H_4 con un exceso de O_2 si la reacción ocurre con un rendimiento del 80%.



* Ver si la reacción está balanceada

3 C 3

4 H 4

8 O 8

∴ La reacción está balanceada

* Calcular ΔH_r° a partir de ΔH_f°

$$\Delta H_f^\circ C_3H_4(g) = +185.4 \frac{kJ}{mol}$$

$$\Delta H_f^\circ O_2(g) = 0$$

$$\Delta H_f^\circ CO_2(g) = -393.5 \frac{kJ}{mol}$$

$$\Delta H_f^\circ H_2O(g) = -241.8 \frac{kJ}{mol}$$

Por lo tanto

$$[3 \text{ mol} (\Delta H_f^\circ CO_2) + 2 \text{ mol} (\Delta H_f^\circ H_2O)] - [(\Delta H_f^\circ C_3H_4) +$$

$$4 \text{ mol} (\Delta H_f^\circ O_2)]$$

$$\Delta H_r^\circ = [3 \text{ mol} (-393.5 \frac{kJ}{mol}) + 2 \text{ mol} (-241.8 \frac{kJ}{mol})] -$$

$$[(185.4 \frac{kJ}{mol}) + 4 \text{ mol} (0)]$$

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$$\Delta H_r^\circ = -1849.5 \text{ KJ}$$

$$325 \text{ g de } C_3H_4 \left(\frac{1 \text{ mol de } C_3H_4}{40.063 \text{ g de } C_3H_4} \right) = 8.112 \text{ mol de } C_3H_4$$

Por lo tanto

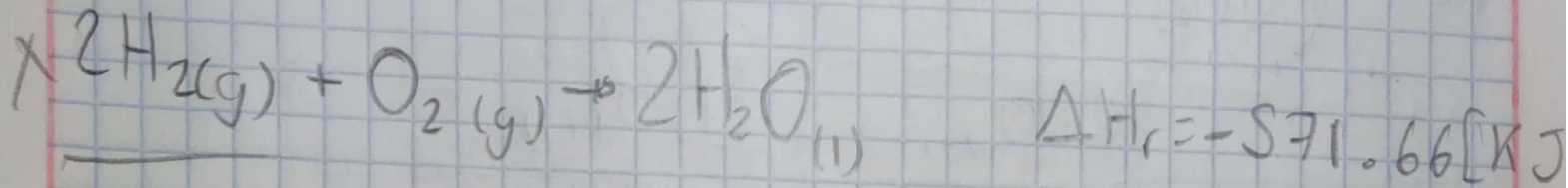
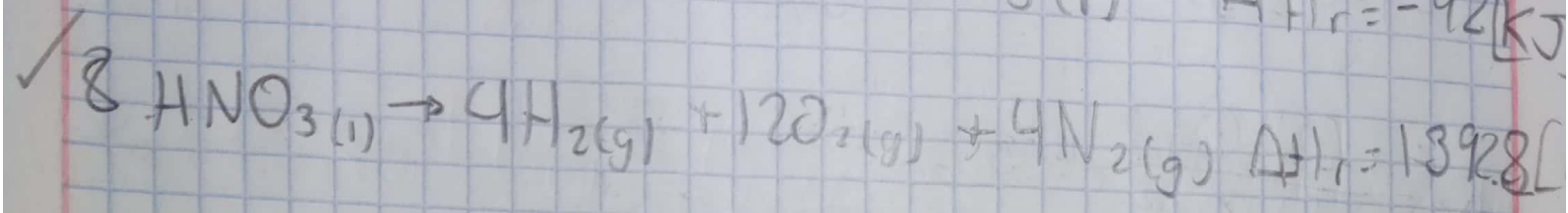
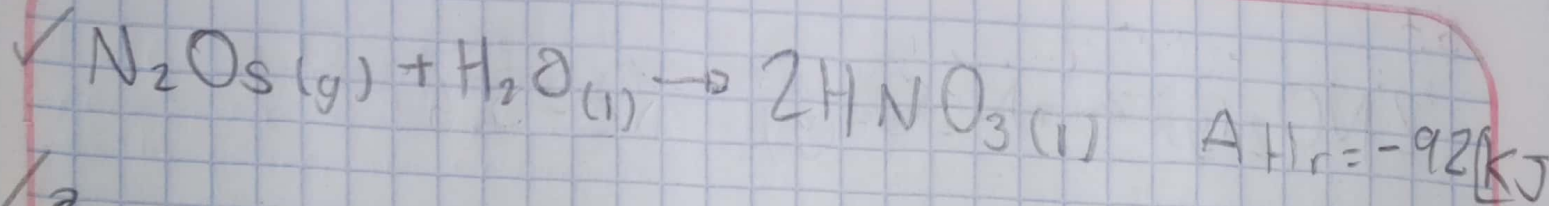
$$\left(\frac{-1849.5 \text{ KJ}}{1 \text{ mol de } C_3H_4} \right) (8.112 \text{ mol de } C_3H_4) = Q_r = 15003.144 \text{ KJ}$$

$$-(1849 \text{ KJ}) \left(\frac{1 \text{ mol}}{(40.063)(325)} \right) = \Delta H = 0.1420 \frac{\text{KJ}}{\text{g}}$$

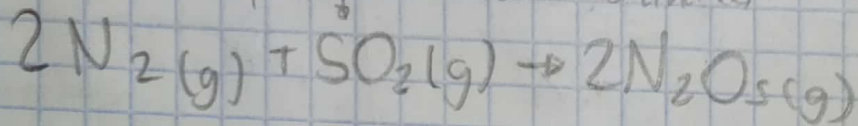
$$0.1420 \times 80 = 11.36$$

$$\text{Rendimiento del } \frac{11.36}{80\%} = 0.1136 \frac{\text{KJ}}{\text{g}}$$

2.-



~~Cate~~ * Comprobar si está balanceada

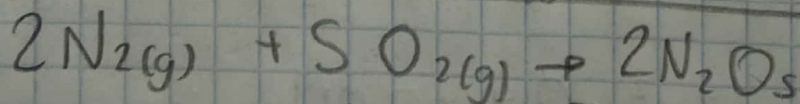
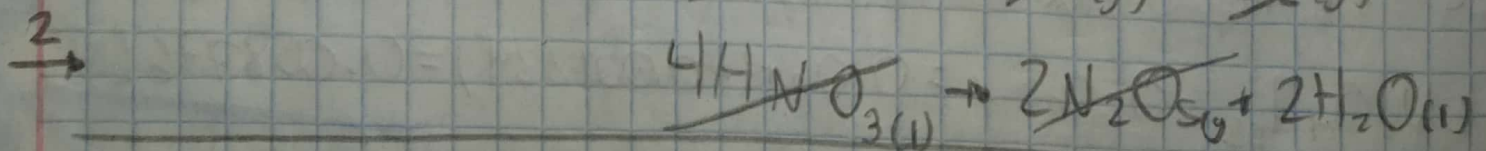
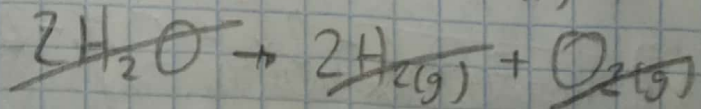
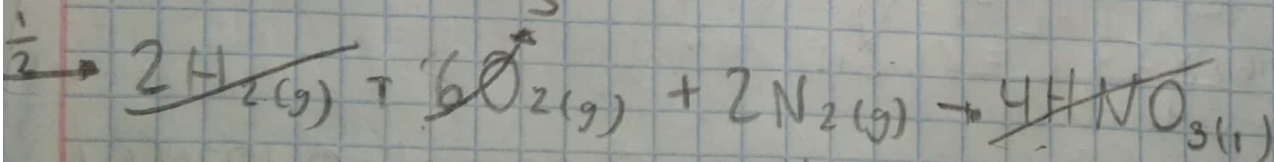


4 N 4

10 O 10

∴ Esta balanceada

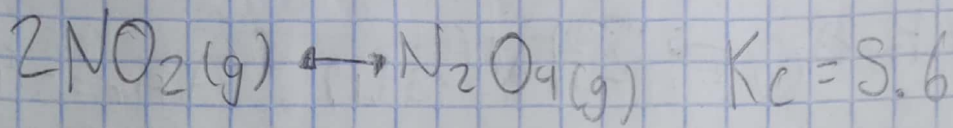
* Se acomodan e invierten



$$\Delta H_r = (+1392.8[\text{KJ}])(-1)\left(\frac{1}{2}\right) \quad \Delta H_r = [-92[\text{KJ}])(-1)(2)$$

$$\Delta H_r = (-571.66[\text{KJ}])(-1) \quad \Delta H_r = 59.26[\text{KJ}]$$

3.-



$$K_c = \frac{[\text{N}_2\text{O}_4]}{[\text{NO}_2]^2} = 5.6$$

	$2\text{NO}_2(\text{g})$	$\text{N}_2\text{O}_4(\text{g})$
Inicio	$\frac{0.1 \text{ M}}{2 \text{ L}}$	$\frac{0.42 \text{ M}}{2 \text{ L}}$
Cambio	$-2x$	$+x$
equilibrio	$0.1 - 2x$	$0.06 + x$

$$K_c = 5.6 = \frac{[0.06 + x]}{[0.1 - 2x]^2} \quad \begin{matrix} x_1 = -0.001224 \\ x_2 = 0.145 \end{matrix}$$

$$5.6 = \frac{0.06 + x}{4x^2 - 0.4x + 0.01}$$

$$[\text{NO}_2]_{\text{eq}} = 0.1 - 2x = 0.1 - 2(-0.001224) = 0.1024 \text{ [M]}$$

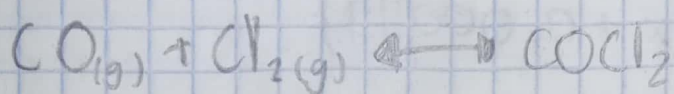
$$[\text{N}_2\text{O}_4]_{\text{eq}} = 0.06 + x = 0.06 + (-0.001224) = 0.058776 \text{ [M]}$$

$$0.06 + x = 5.6(4x^2 - 0.4x + 0.01)$$

$$x = 22.4x^2 - 2.24x + 0.056 - 0.06$$

$$x = 22.4x^2 - 2.24x - 0.004$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad x = \frac{-(-2.24) \pm \sqrt{(-2.24)^2 - 4(22.4)(-0.004)}}{2(22.4)}$$



$$K_c = 1.5 \times 10^4 = \frac{[\text{COCl}_2]}{[\text{CO(g)}][\text{Cl}_2\text{(g)}]}$$

$$140 \text{ L} = 140 \text{ L}$$

	CO(g)	$\text{Cl}_2\text{(g)}$	$\text{COCl}_2\text{(g)}$
Inicio	$\frac{2.499 \text{ mol}}{140 \text{ L}}$	$\frac{0.989 \text{ mol}}{140 \text{ L}}$	0
Cambio	-x	-x	+x
Equilibrio	$0.017 - x$	$0.007 - x$	$0 + x$

$$70 \text{ g de Cl}_2 \left(\frac{1 \text{ mol de Cl}_2}{70.91 \text{ g de Cl}_2} \right) = 0.989 \text{ [mol] de Cl}_2$$

$$70 \text{ g de CO} \left(\frac{1 \text{ mol de CO}}{28.01 \text{ g de CO}} \right) = 2.499 \text{ [mol] de CO}$$

$$K_c = 1.5 \times 10^4 = \frac{[x]}{[0.017 - x][0.007 - x]}$$

$$1.5 \times 10^4 = \frac{x}{x^2 - 0.024x + 0.000119}$$

$$0.006953$$

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$$x = 1.5 \times 10^4 (0.024x + 0.000119)$$

$$x = 15000x^2 - 360x + 1785$$

$$x = \frac{-(-360) \pm \sqrt{(-360)^2 - 4(15000)(1785)}}{2(15000)}$$

$$x = \frac{360 \pm \sqrt{129600 - 107100000}}{30000}$$

$$x = \frac{360 \pm 10342.649}{30000}$$

$$x_1 = 0.006953 \quad x_2 = 0.017112$$

$$[CO]_e = 0.017 - x = 0.010047$$

$$[Cl_2]_e = 0.007 - x = 4.7 \times 10^{-5}$$

$$[COCl_2]_e = x = 0.006953$$